Gantt Charts

Planning a software project with Excel

1. What is a Gantt chart?

A Gantt chart is a **timeline drawn as bars**. It is one of the main tools of **project planning** in software engineering. It answers three questions:

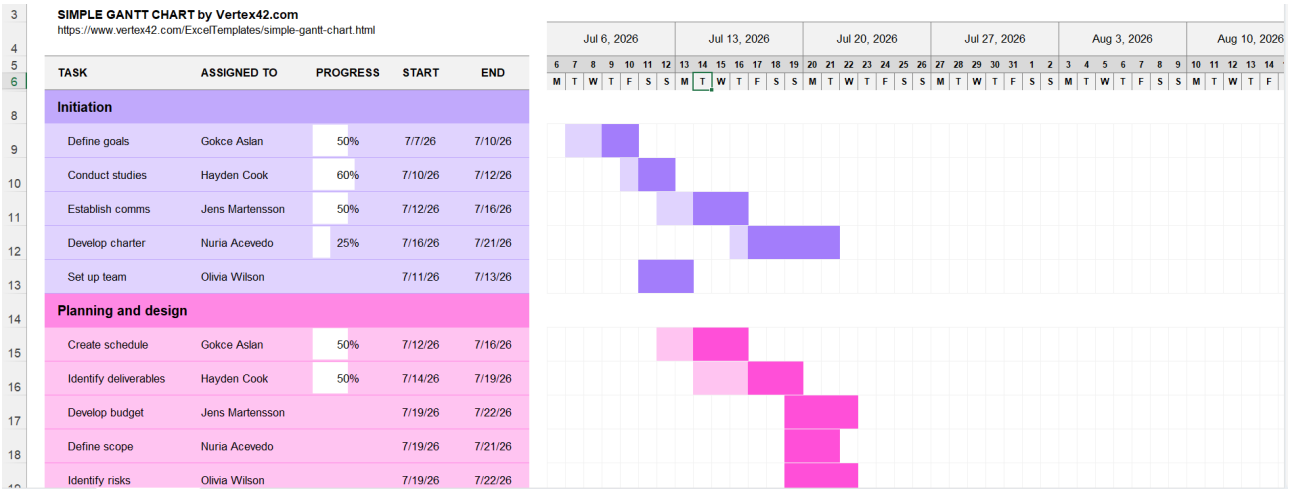
- **What** tasks must be done?
- **When** does each task start and end?
- **Which tasks** must finish before others can start?

2. Key terms

Term	Meaning	Example
Task	A unit of work	Design the database
Start / End	When a task begins and finishes	7/7/26 to 7/10/26
Duration	Number of days a task takes	End - Start
Dependency	Task B cannot start until task A finishes	Testing after Coding
Milestone	An important checkpoint	First version delivered
Progress %	How much of the task is done	50%
Critical path	Longest chain of dependent tasks. A delay here delays the whole project	Req. → Design → Code → Test

3. How to read the chart

The **left side** is the task table (Task, Assigned to, Progress, Start, End). The **right side** is the timeline: each colored bar shows when a task is active. The darker part of a bar is the completed percentage.



SIMPLE GANTT CHART by Vertex42.com
https://www.vertex42.com/ExcelTemplates/simple-gantt-chart.html

TASK	ASSIGNED TO	PROGRESS	START	END
Initiation				
Define goals	Gokce Aslan	50%	7/7/26	7/10/26
Conduct studies	Hayden Cook	60%	7/10/26	7/12/26
Establish comms	Jens Martensson	50%	7/12/26	7/16/26
Develop charter	Nuria Acevedo	25%	7/16/26	7/21/26
Set up team	Olivia Wilson		7/11/26	7/13/26
Planning and design				
Create schedule	Gokce Aslan	50%	7/12/26	7/16/26
Identify deliverables	Hayden Cook	50%	7/14/26	7/19/26
Develop budget	Jens Martensson		7/19/26	7/22/26
Define scope	Nuria Acevedo		7/19/26	7/21/26
Identify risks	Olivia Wilson		7/19/26	7/22/26

Figure 1: Simple Gantt chart template in Excel (Europa project sample)

Look at the picture:

- **Stair-step bars** (Define goals → Conduct studies → Establish comms → Develop charter): each task starts when the previous one ends. These are **dependent** tasks.
- **“Set up team”** starts on 7/11, so it runs **at the same time** as other tasks. This is **parallel** work.
- **Initiation** (purple) and **Planning and design** (pink) are two **phases**. They overlap in time.

4. Sequential vs. parallel tasks

Not all tasks run at the same time. It depends on **dependencies**.

	Sequential (dependent)	Parallel (independent)
Question to ask	Can I start without the previous task finishing? **No**	Can I start without the previous task finishing? **Yes**
Start date	= End date of previous task	Its own date
Example	Coding → Testing	Coding + writing test cases

Fast tracking: running tasks or phases in parallel (overlap) to shorten the project. It saves time, but increases risk if work starts before earlier decisions are final.

5. Critical path

The **critical path** is the longest chain of dependent tasks. It decides the total project duration. Parallel tasks save time, but if a task on the critical path is late, the **whole project is late**.

6. Using the Excel template

In this template, the **END** column is typed by hand and the **START** column is a formula that takes the END of the previous task.

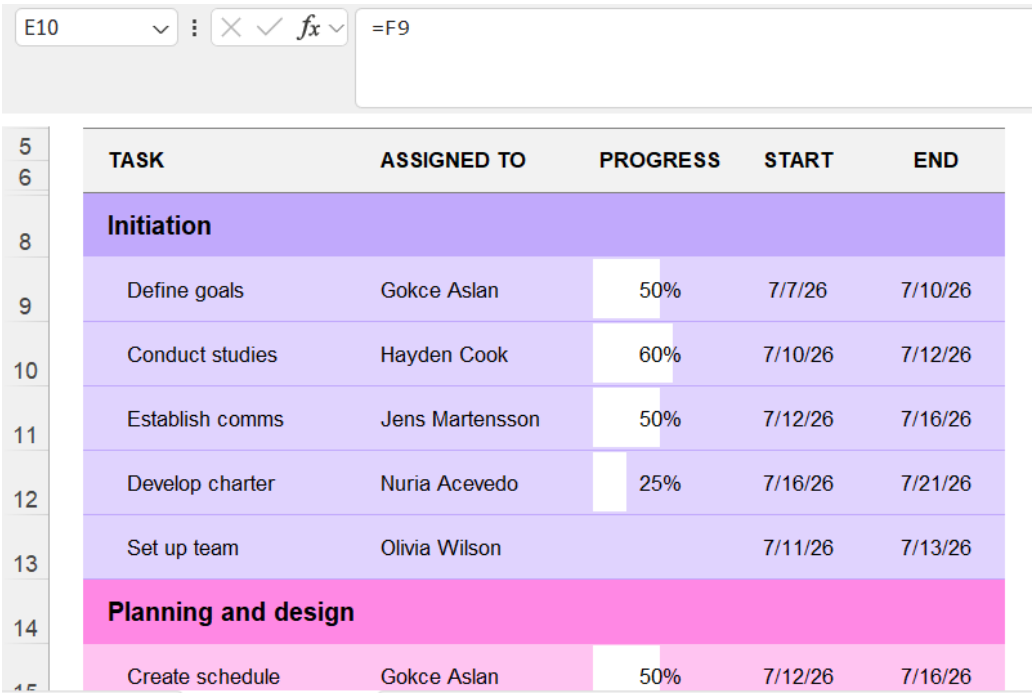


Figure 2: Cell E10 (Start of "Conduct studies") contains the formula =F9, the End date of "Define goals"

What to change

- **Project start** (top of the sheet): moves the whole project.
- **END** of a task: all tasks after it shift automatically.
- **Task name**, **Assigned to**, **Progress %**: type freely.
- **START** only for an independent task (for example "Set up team") that has no previous task.

Rules to avoid breaking the sheet

- Click a cell and check the **formula bar**. If it starts with **=**, do not type over it.
- Save a copy first (**File > Save As**).
- Change one thing at a time. Use **Ctrl+Z** to undo.
- Do not touch the date row above the colored bars. It is calculated.
- Use **Display week** to scroll the timeline if the bars disappear.

7. Worked problem: Gantt chart from a precedence table

A project has 10 activities (A to J). The duration of each activity and the activities that must finish before it can start (precedence) are given. **Draw a Gantt chart and find the project completion time.**

Activity	A	B	C	D	E	F	G	H	I	J
Precedence	-	-	A, B	A	C	A, B	F	G	G	I
Duration (days)	40	20	15	25	15	25	10	15	10	15

Rule: every activity starts as early as possible

- **No predecessor:** Start = 0
- **One predecessor:** Start = End of that predecessor
- **Several predecessors:** Start = the **largest** End among them (it must wait for the last one)
- **End = Start + Duration**

Solution table

Activity	Duration	Start	End	Why this Start?
A	40	0	40	no predecessor
B	20	0	20	no predecessor
C	15	40	55	max(End A = 40, End B = 20) = 40
D	25	40	65	End A = 40
E	15	55	70	End C = 55
F	25	40	65	max(40, 20) = 40
G	10	65	75	End F = 65
H	15	75	90	End G = 75
I	10	75	85	End G = 75
J	15	85	100	End I = 85

Project completion time = 100 days (activity J ends at day 100).

Gantt chart of the solution

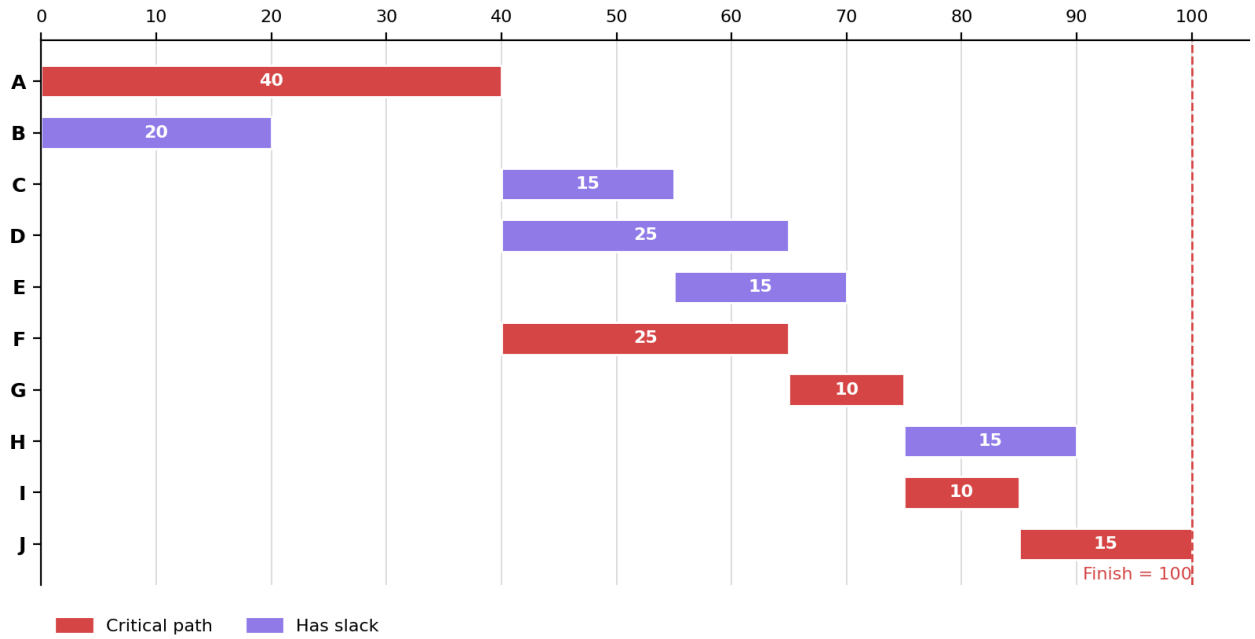


Figure 3: Each bar starts at its Start day and its length is the Duration

Critical path

The critical path is the chain with no spare time: **A** → **F** → **G** → **I** → **J** (40 + 25 + 10 + 10 + 15 = 100 days). A delay in any of these delays the whole project.

The other activities have **slack** (time they can slip without delaying the project):

Activity	B	C	D	E	H
Slack (days)	20	30	35	30	10

Note: the original lecture slide draws the time axis from 120 down to 0 and calls the Start column “precedence”. Here the axis runs normally from 0 to 100 and the column is called Start.

9. Building it in Excel

Step 1. Type the table: columns **Activity, Predecessors, Duration, Start, End** (rows 5 to 14 for A to J).

Step 2. Type these formulas in the **Start** column (D):

Activity	Start formula (column D)	Meaning
A, B	0	no predecessor
C	=MAX(E5,E6)	last of A and B to finish
D	=E5	after A
E	=E7	after C
F	=MAX(E5,E6)	last of A and B to finish
G	=E10	after F
H, I	=E11	after G
J	=E13	after I

Step 3. In the **End** column (E) type **=D5+C5** and drag it down to row 14.

Step 4. In any cell type **=MAX(E5:E14)**. The answer is **100**.

Step 5. Draw the chart:

- Select the **Activity** column, then hold **Ctrl** and select **Start** and **Duration**.
- **Insert > Bar chart > Stacked Bar**.
- Click the **Start** bars > **Format Data Series > Fill: No fill** (they disappear; only the Duration bars remain).
- Click the activity axis (A, B, C...) > **Format Axis > Categories in reverse order** so that A is on top.

Now change any Duration and watch Start, End, the chart and the total update. This shows how dependencies work.

8. Class exercise

Build a Gantt chart for a **course registration website**. Start on day 1.

Task	Days	Depends on
Gather requirements	5	-
Design database	4	Gather requirements
Design user interface	5	Gather requirements
Coding	15	Both designs
Testing	7	Coding
Delivery	1	Testing

Questions

- Which tasks can run in parallel?
- What is the critical path?
- What happens to the delivery date if Coding takes 20 days instead of 15?
- Which task would you shorten to finish the project sooner?